Introduction to Vectors

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Q1

The slopes are v_2/v_1 and w_2/w_1. The hypothesis gives

$$\frac{v_2}{v_1}\frac{w_2}{w_1} = -1.$$

Since $v_1 w_1 \neq 0$, multiplication by $v_1 w_1$ is valid and yields

$$v_2 w_2 = -v_1 w_1.$$

Therefore

$$v \cdot w = v_1 w_1 + v_2 w_2 = 0.$$

By the definition of orthogonality, v and w are perpendicular.

Q2

(i) For each i, let $r_i \in \mathbb{R}^n$ be a column vector such that r_i^T is the ith row of A. Writing A row by row gives

$$A = \begin{bmatrix} r_1^T \\ \vdots \\ r_m^T \end{bmatrix}.$$

Therefore

$$Ax = \begin{bmatrix} r_1^T x \\ \vdots \\ r_m^T x \end{bmatrix} = \begin{bmatrix} r_1 \cdot x \\ \vdots \\ r_m \cdot x \end{bmatrix}.$$

If $Ax = 0$, every component of this vector is zero, so

$$r_i \cdot x = 0 \qquad (i = 1, \ldots, m).$$

Thus every row is perpendicular to x.

(ii) Any linear combination of the rows is also perpendicular to x, because

$$\left(\sum_i c_i r_i \right) \cdot x = \sum_i c_i (r_i \cdot x) = 0.$$

If the rows span a plane, every vector in that plane is perpendicular to x. Hence the plane is perpendicular to x.

Q3

(i) The equations in $Ax = b$ are

$$x_1 = b_1, \qquad -x_1 + x_2 = b_2, \qquad x_1 - x_2 + x_3 = b_3.$$

They can be solved from top to bottom:

$$x_1 = b_1, \qquad x_2 = b_1 + b_2, \qquad x_3 = b_2 + b_3.$$

Thus

$$x = \begin{bmatrix} b_1 \\ b_1 + b_2 \\ b_2 + b_3 \end{bmatrix}.$$

(ii) Writing the preceding solution as $x = A^{-1}b$ gives

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Q4

(i) We seek c, d, e such that

$$c(1,3) + d(2,7) + e(1,5) = (0,1).$$

Equating components gives

$$c + 2d + e = 0, \qquad 3c + 7d + 5e = 1.$$

There are two free choices. Taking $e = 0$ gives $c = -2d$ and then $d = 1$, so one combination is

$$-2u + v = b.$$

Taking $d = 0$ gives $c = -e$ and $2e = 1$, so another is

$$-\frac{1}{2}u + \frac{1}{2}w = b.$$

(ii) For the general statement, three vectors in the plane are linearly dependent. Thus there are scalars α, β, γ, not all zero, such that

$$\alpha u + \beta v + \gamma w = 0.$$

If (c_0, d_0, e_0) produces b, then for every real t,

$$(c_0 + t\alpha)u + (d_0 + t\beta)v + (e_0 + t\gamma)w = b + t0 = b.$$

Distinct values of t give infinitely many distinct coefficient triples.

Q5

Because x, y, z are not all zero, $v \neq 0$; then $w \neq 0$ because w is a permutation of the components of v. We have

$$v \cdot w = xz + xy + yz.$$

Squaring $x + y + z = 0$ gives

$$x^2 + y^2 + z^2 + 2(xy + yz + zx) = 0,$$

so

$$v \cdot w = -\frac{1}{2}(x^2 + y^2 + z^2).$$

Since w is a permutation of v,

$$\|v\|^2 = \|w\|^2 = x^2 + y^2 + z^2.$$

Consequently,

$$\frac{v \cdot w}{\|v\|\|w\|} = -\frac{1}{2}.$$

Thus the angle is

$$\theta = \arccos\left(-\frac{1}{2}\right) = 120^\circ.$$

Q6

Suppose first that the rows of A are dependent. If the second row is zero, then $c = d = 0$ and the columns are

$$\begin{bmatrix} a \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} b \\ 0 \end{bmatrix},$$

which are both multiples of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Hence they are dependent.

Otherwise, there is a scalar λ such that

$$(a, b) = \lambda(c, d).$$

The columns can then be written as

$$\begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} \lambda c \\ c \end{bmatrix} = c \begin{bmatrix} \lambda \\ 1 \end{bmatrix}, \qquad \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} \lambda d \\ d \end{bmatrix} = d \begin{bmatrix} \lambda \\ 1 \end{bmatrix}.$$

Thus both columns are scalar multiples of the same vector, so the columns are dependent. No division by a matrix entry is required.

Conversely, suppose the columns of A are dependent. Those columns are the rows of A^T. Applying the implication just proved to A^T shows that the columns of A^T are dependent. Those columns are precisely the rows of A. Hence the rows are dependent, completing the proof.

Q7

(i) Suppose first that u and v are nonzero orthogonal vectors in $\mathbb{R}^2$. They point in two independent directions and therefore span the plane. If a vector w is orthogonal to both, write

$$w = \alpha u + \beta v.$$

Taking dot products with u and v gives

$$w \cdot u = \alpha \|u\|^2 = 0, \qquad w \cdot v = \beta \|v\|^2 = 0.$$

Since u and v are nonzero, $\alpha = \beta = 0$, so $w = 0$. Thus a third vector orthogonal to both u and v must be zero. Therefore three nonzero mutually orthogonal vectors cannot exist in $\mathbb{R}^2$.

(ii) Let $v_1, \ldots, v_k$ be nonzero mutually orthogonal vectors in $\mathbb{R}^n$. We first show that they are linearly independent. If

$$c_1 v_1 + \cdots + c_k v_k = 0,$$

then taking the dot product with v_j gives

$$c_j \|v_j\|^2 = 0.$$

Because $v_j \neq 0$, this implies $c_j = 0$. The same argument applies for every j, so all coefficients are zero. Hence $v_1, \ldots, v_k$ are linearly independent. Since an n-dimensional space has at most n linearly independent vectors, $k \leq n$.

(iii) *Optional.* Let $M_n(\varepsilon)$ denote the largest number of unit vectors with all pairwise angles in $(90^\circ - \varepsilon, 90^\circ)$. A plausible conjecture is that, for each fixed $\varepsilon > 0$, $M_n(\varepsilon)$ grows exponentially with n. The motivation is that high-dimensional spaces contain many directions that are almost orthogonal. The exact orthogonality condition permits only n nonzero vectors, by part (ii), but allowing a fixed small departure from 90° should permit substantially more vectors. Nevertheless, establishing a sharp formula belongs to the theory of spherical codes and is beyond the scope of this worksheet.

Q8

(i) Rotational symmetry allows us to fix $u = (1, 0)$. If θ is the angle from u to v, then $u \cdot v = \cos\theta$. The mean absolute dot product is therefore

$$\mathbb{E}|u \cdot v| = \frac{1}{\pi} \int_0^\pi |\cos\theta| \, d\theta = \frac{2}{\pi}.$$

Thus the mean absolute dot product in $\mathbb{R}^2$ is $\boxed{2/\pi}$.

(ii) *Optional.* In $\mathbb{R}^3$, rotational symmetry allows us to fix $u = (0, 0, 1)$. A uniformly distributed point on the unit sphere has surface area element $\sin\theta \, d\theta \, d\phi$. A horizontal spherical band between heights z and $z + dz$ has area $2\pi \, dz$, independent of z. Since the total sphere has area 4π, the random variable

$$Z = u \cdot v$$

is uniformly distributed on $[-1, 1]$, with density $f_Z(z) = \frac{1}{2}$.
Using the expectation formula from the hint,

$$\mathbb{E}|u \cdot v| = \mathbb{E}|Z| = \int_{-1}^{1} |z| \frac{1}{2} \, dz = \frac{1}{2}.$$

Thus the mean absolute dot product is $\boxed{\frac{1}{2}}$. Because $1/2 < 2/\pi$, randomly chosen unit vectors are, on average, closer to right angles in $\mathbb{R}^3$ than in $\mathbb{R}^2$. This supports the intuition behind Q7(iii), although it does not prove the conjecture: Q7(iii) additionally requires every pair in a large collection to form an acute angle.